Abstract

Exploratory data analysis (EDA) is the most common entry point in applied research, yet it is also where reproducibility most often breaks down: logic accumulates in notebook cells that are never tested and quietly drift from the prose describing them. This paper presents the **computational-notebook exemplar** of the Research Project Template: an interactive walkthrough notebook (`projects/templates/template_eda_notebook/notebooks/eda_walkthrough.ipynb`) that imports a small, fully-tested EDA library rather than carrying logic in its cells.

We ship a deterministic dataset (`data/measurements.csv`) with a designed correlation structure and a handful of missing values, then load, clean, summarize, correlate, and visualize it entirely through tested functions in `src/eda/`. The library is side-effect-free — no plotting and no file I/O — and standalone (numpy and pandas only), so it is covered above the 90% project gate and reused identically from the notebook, the thin analysis script (`scripts/eda_analysis.py`), and this manuscript.

Abstract I

Contributions are **methodological** and **architectural**. On the methods side, we walk the canonical first EDA pass: surface missingness explicitly rather than imputing it, compute per-column descriptive statistics and per-group means, and rank features by Pearson correlation. On the architecture side, we demonstrate the notebook-to-tested-source extraction workflow — explore fast in a cell, and the moment a computation matters, move it into the library behind a failing test — verified by a zero-mock suite and a structural notebook-binding check (sec. 7).